

FloodGuard Thailand — THEOS-2 Imagery Request

Request reference: FG-THEOS2-REQ-001 **Date:** 17 September 2026 **Submitted to:** GISTDA, THEOS-2 imagery request **Contact:** [name], [institution / affiliation], [email]

1. About the project

FloodGuard Thailand is a non-operational research and preparedness prototype. It converts flood extent or flood probability into subdistrict-level decision support: exposed population, likely road disruption, evacuation access loss, an equity gap measure, and a flood preparedness priority score.

The project does not publish official warnings, does not claim real-time flood detection, and does not redistribute source imagery. It is not a replacement for GISTDA, DDPM, TMD, RID, ONWR, or local agency judgement.

Data handling. Any imagery provided would be held in a controlled workspace outside version control, tracked by SHA-256 checksum. Only derived analytical outputs are retained in the project repository. Source imagery is never redistributed. Citation and attribution will follow whatever form GISTDA specifies.

2. Areas of interest

Six areas across three flood events. All bounds are WGS84 (EPSG:4326), given as `min_lon`, `min_lat`, `max_lon`, `max_lat`. Scene counts assume a ~10.3 km THEOS-2 PMS footprint.

Accompanying file: `floodguard_theos2_aoi_v1.geojson` (all six AOIs). Individual per-AOI files are also provided.

AOI	Priority	Area	Province / District	Bounding box (WGS84)	Size	~Scenes
AOI-01	1	Mae Sai core — town, border crossing, Sai River	Chiang Rai / Mae Sai	99.8384, 20.3705, 99.9441, 20.4563	11.0 × 9.5 km	~2
AOI-02	2	Mae Sai district — all 8 subdistricts	Chiang Rai / Mae Sai	99.8107, 20.2577, 100.0368, 20.4651	23.6 × 22.9 km	~9
AOI-03	1	Hat Yai core — municipality, U Taphao canal	Songkhla / Hat Yai	100.4300, 6.9650, 100.5200, 7.0550	9.9 × 10.0 km	~1
AOI-04	3	U Taphao basin — extended	Songkhla	100.3800, 6.9000, 100.5800, 7.1200	22.1 × 24.3 km	~9

AOI	Priority	Area	Province / District	Bounding box (WGS84)	Size	~Scenes
AOI-05	1	Bang Ban / Sena — lower Chao Phraya	Ayutthaya	100.4000, 14.2500, 100.5200, 14.3700	12.9 × 13.3 km	~4
AOI-06	2	Rangsit / Thanyaburi	Pathum Thani	100.6000, 13.9800, 100.7200, 14.0800	13.0 × 11.1 km	~4

3. Requested acquisitions

Date ranges were selected by measurement, not estimation. For every area and candidate period we queried Sentinel-2 archive imagery and computed the cloud fraction **clipped to the AOI polygon**, rather than relying on scene-level cloud percentage. The ranges below are narrowed around dates that were demonstrably clear over the target area.

Two cloud figures are given, and they are not interchangeable. "Cloud over our area" is the measured cloud fraction within the AOI polygon, which is what determines whether an image is usable to us. "Portal filter" is the scene-level value we believe should be entered in the ordering portal so that the dates we want are actually returned. The note "Why two cloud figures" below explains the difference.

Areas	Date range	Cloud over our area	Portal filter	Purpose	Most promising dates
AOI-01, 02	2024-09-01 → 2024-09-08	40%	40%	Immediate pre-flood baseline	2024-09-05
AOI-01, 02	2024-09-09 → 2024-09-30	95%	95%	Flood extent (opportunistic)	—
AOI-01, 02	2025-01-05 → 2025-01-31	10%	45%	Clear reference baseline	2025-01-13, 01-18, 01-23
AOI-01, 02	2024-11-01 → 2024-11-20	10%	40%	Early post-event baseline	2024-11-04, 11-14
AOI-03, 04	2025-11-28 → 2025-12-08	70%	70%	Flood extent	2025-12-03 , 12-01

Areas	Date range	Cloud over our area	Portal filter	Purpose	Most promising dates
AOI-03, 04	2025-03-25 → 2025-04-05	15%	25%	Pre-event baseline	2025-03-31
AOI-03, 04	2026-03-15 → 2026-04-15	10%	30%	Clear reference baseline	2026-03-21, 03-26, 04-12
AOI-05, 06	2024-10-07 → 2024-11-15	25%	75%	Flood extent, 2024	2024-11-11, 2024-10-12
AOI-05, 06	2025-10-15 → 2025-11-20	70%	90%	Flood extent, 2025	2025-11-16
AOI-05, 06	2025-01-15 → 2025-02-20	10%	55%	Dry-season baseline	2025-02-04, 02-14, 02-19

If a single filter value is easier to apply, 95% is safe for every window above. Because we have supplied specific target dates, selecting by date rather than by cloud threshold will give the most reliable result.

Preferred product

ORTHO PMS — pan-sharpened multispectral, 4-band (Blue / Green / Red / NIR).

This matches the IMG_T2V_*_ORTHO_PMS_* samples already available to us from the hackathon dataset. The near-infrared band is important: it is what makes optical water discrimination possible and allows us to screen false positives in our Sentinel-1 radar analysis. Panchromatic-only imagery would be considerably less useful for this work.

Why two cloud figures

Our areas of interest are 10–24 km across; a satellite scene covers a far larger footprint. As a result, scene-level cloud cover and the cloud actually sitting over our area diverge substantially — in both directions:

Date	Area	Scene-level cloud	Cloud over our area
2024-11-21	AOI-06, Pathum Thani	62.5%	0.8%
2025-02-04	AOI-06, Pathum Thani	51.0%	0.1%
2025-02-19	AOI-06, Pathum Thani	46.3%	0.1%

Date	Area	Scene-level cloud	Cloud over our area
2025-01-28	AOI-01, Mae Sai	41.9%	0.3%
2024-10-12	AOI-05, Ayutthaya	60.5%	11.2%
2025-12-03	AOI-03, Hat Yai	27.9%	39.3% ← reverse case

The practical consequence: if the cloud figure we care about were entered directly into a scene-level filter, **17 of the dates we are asking for would be excluded**. The clearest example is 2025-02-04 and 2025-02-19 over Pathum Thani — essentially cloud-free over our area at 0.1%, but around 50% at scene level, so a 10% filter would reject both.

This is why the table above gives a separate, higher "portal filter" value. Those values are calculated to be the lowest scene-level threshold that still returns every date we want.

Note on the Mae Sai flood window

We have set the cloud limit to 95% for the September 2024 Mae Sai flood window deliberately. Our measurements show the flood peak was completely overcast (2024-09-10 measured 100% cloud over the area), so we expect nothing there and a nil result is entirely acceptable.

The acquisition of 2024-09-05, five days before the peak, is considerably more valuable to us — it provides a seasonally matched pre-flood baseline. For Hat Yai and the lower Chao Phraya, by contrast, our measurements suggest that usable imagery during the flood period may genuinely exist.

If capacity is limited

Our four highest-value acquisitions, in order:

- 1. **AOI-01, January 2025** (13th, 18th or 23rd) — clear imagery of our primary study area.
- 2. **AOI-01, 2024-09-05** — the pre-flood baseline for Mae Sai.
- 3. **AOI-05, 2024-11-11 or 2024-10-12** — imagery during the Chao Phraya flood period.
- 4. **AOI-03, 2025-12-03** — partial imagery during the Hat Yai flood.

4. Additional data request

Alongside the imagery, the following datasets would substantially strengthen the work. We understand that some may sit with other agencies, or may not be shareable. Any pointer to the appropriate contact would be very welcome.

Priority	Dataset	Period required
1	Official flood extent / flood map products	Mae Sai: 2024-09-09 → 2024-09-25 Hat Yai: 2025-11-19 → 2025-12-15 Chao Phraya: 2024-09-25 → 2024-11-30 and 2025-09-20 → 2025-11-30
2	Official shelter / evacuation-centre locations (DDPM or provincial)	Current register, plus the register as it stood in September 2024 and November 2025 if historical versions are retained
3	GISTDA Sentinel-1 derived water-extent products	Event periods as above, plus pre-event reference: Mae Sai 2024-08-15 → 2024-09-08; Hat Yai 2025-10-15 → 2025-11-18; Chao Phraya 2024-08-01 → 2024-09-24
4	Flood hazard or recurrence-zonation maps	Latest vintage; a series from 2011 onward if available
5	Water-level and rainfall station data (RID / TMD)	Mae Sai: 2024-08-01 → 2024-10-15 Hat Yai: 2025-10-01 → 2026-01-15 Chao Phraya: 2024-08-01 → 2024-12-31 and 2025-08-01 → 2025-12-31 Hourly resolution or finer if held
6	Official subdistrict (ADM3) boundaries	Version valid as at 2024-09-01, and the current 2026 version
7	Field-verified flood observations, high-water marks, survey points	Mae Sai: 2024-09-10 → 2024-12-31 Hat Yai: 2025-11-19 → 2026-03-31 Chao Phraya: 2024-10-01 → 2025-01-31
8	Building footprints or official subdistrict population	Latest available vintage, plus any 2024 or 2025 update

Licensing questions

For any flood extent product, we would need written confirmation of the following. Our project applies a strict data-governance rule: any dataset whose terms are unresolved is blocked from entering the analysis, so we are unable to proceed on an informal basis even where permission is likely.

1. Whether the geometry or raster may be used for local research validation.
2. Whether derived validation metrics may be published.
3. Whether it may be used as training labels for a machine-learning model.
4. Whether the source product may be redistributed, or must remain reference-only.
5. Required citation, attribution, and any disclaimers.

We had previously written to GISTDA on 3 July 2026 with a similar enquiry. If that request is better directed elsewhere, we would be grateful for a pointer to the right team.

5. Accompanying files

File	Contents
floodguard_theos2_aoi_v1.geojson	All six areas of interest, WGS84, with priority, purpose and requested windows in the feature properties
aoi-01_mae_sai_core.geojson ... aoi-06_chao_phraya_rangsit.geojson	Individual area files, if one AOI per request is preferred

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